

Did the model really *need* the corpus?

A retrieval-augmented system answers correctly. Was the retrieved knowledge doing the work — or did the model already know? We measure it, and we prove the measure works *by construction*.

1 ABLATE

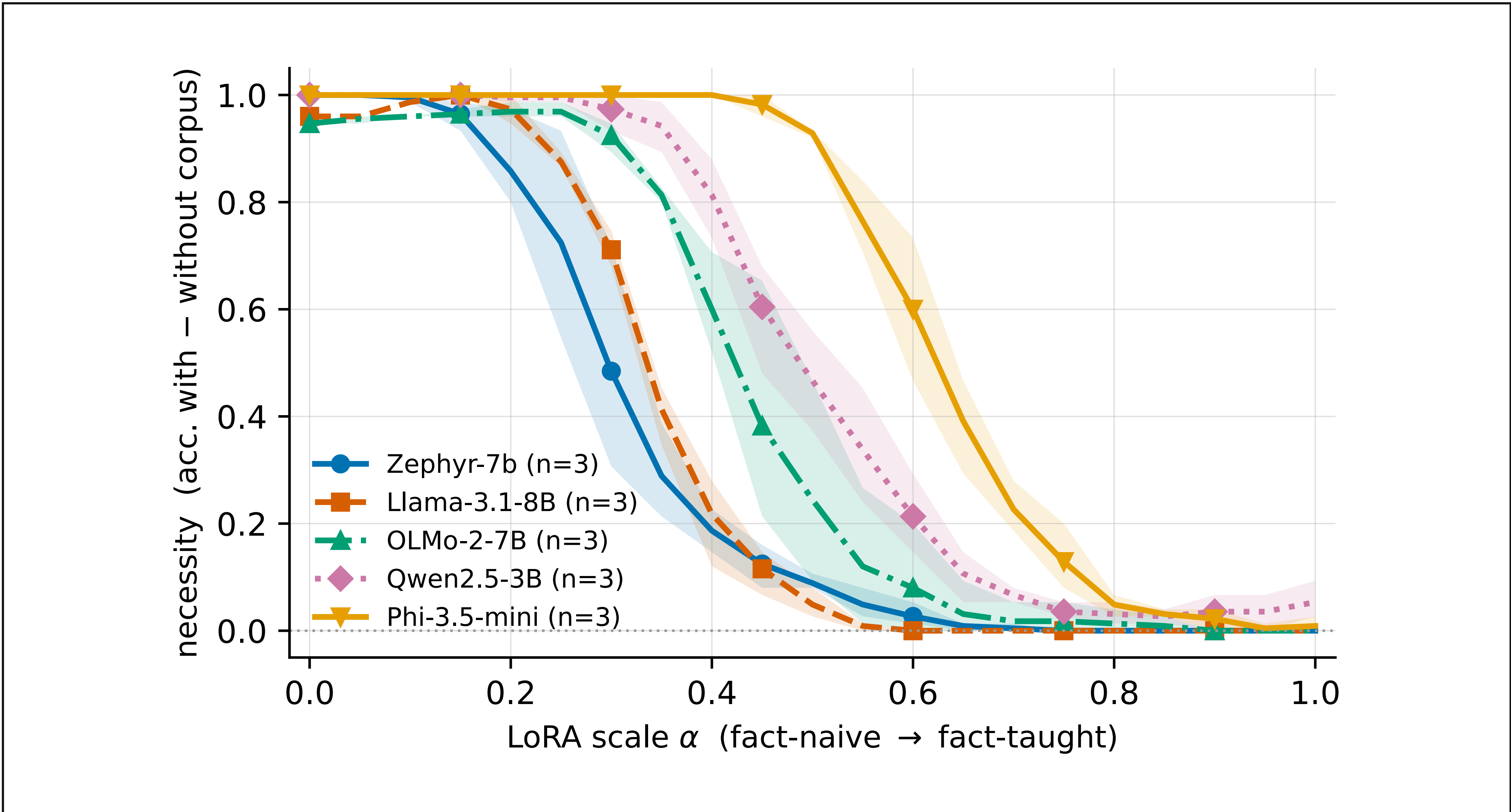
Remove one corpus item and re-run the task.

2 SCORE

Necessity = performance with – without, on a *scored task outcome* — not an influence score.

3 CALIBRATE

Teach invented facts into the weights. If necessity does not fall, the measure is broken. It falls.



The calibration, measured. Five base-model lineages $\times$ three seeds $\times$ a 21-point teaching gradient: necessity collapses $1.00 \rightarrow \leq 0.05$ in every family, each with its transition at a different point — not an artifact of any one model. Every value traces to a committed audit log.

Knowing $\neq$ doing

Teach a ship’s autopilot a hidden hazard *as prose*: it can now **say** the fact (recall $0.00 \rightarrow 0.78$) — but its scored steering decision does not move. Only *reason-then-decide* pulls the knowledge into the action.

naive · single-shot 667 grounds	taught · single-shot 667 knows, doesn’t do
naive · reason first 669.5 grounds	taught · reason first 0.3 clears — gap closed

The misled blind spot

A learner faithfully following a **wrong** retrieved rule looks *maximally reliant* — remove the rule and its decision swings hard — yet it is actively harmed. Influence and faithfulness metrics cannot see this case.

Reading necessity as **decision-changes $\times$ succeeds-with** separates real reliance from faithful compliance with a harmful rule — the failure this instrument exists to expose.

Live-model demonstration: a wrong rule made the harmful maneuver 8 $\times$ more likely — and the model explained why it was safe.

Paper · plain-language companion · full pipeline · per-fact audit logs

Every number in the paper traces to a committed file. Offline self-check reproduces the ground truth with no GPU and no API key:
`npm run reproduce`

Simulation-based mechanism evidence — the pre-human-study screen, honestly scoped.



github.com/wrgr/socratic-scenarios